

Homework 7

1. Consider graphene.

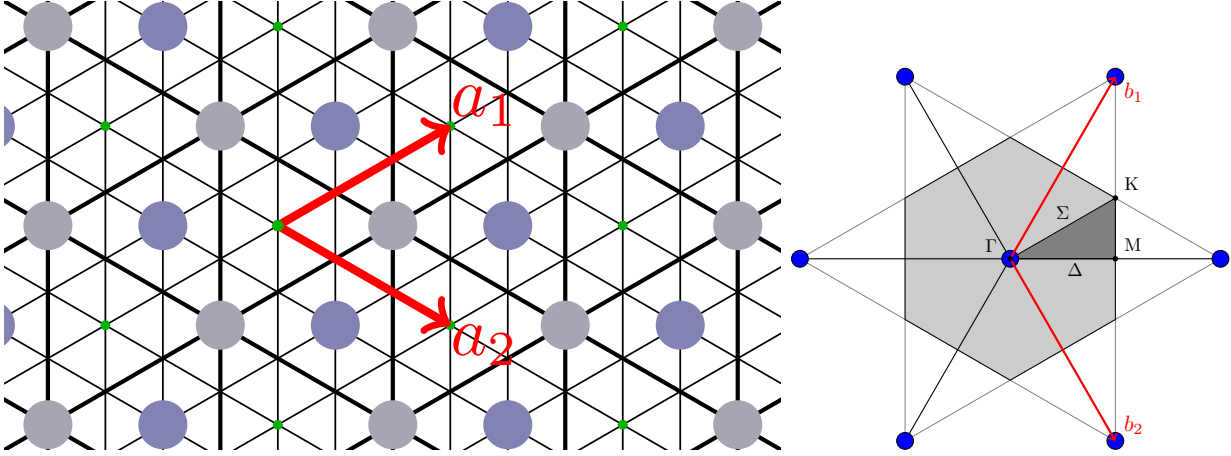


Figure 1: (Left panel) The structure of graphene. (Right Panel) The first Brillouin Zone and the irreducible Brillouin zone for the triangular lattice.

Now consider the potential, which we will first schematically illustrate:

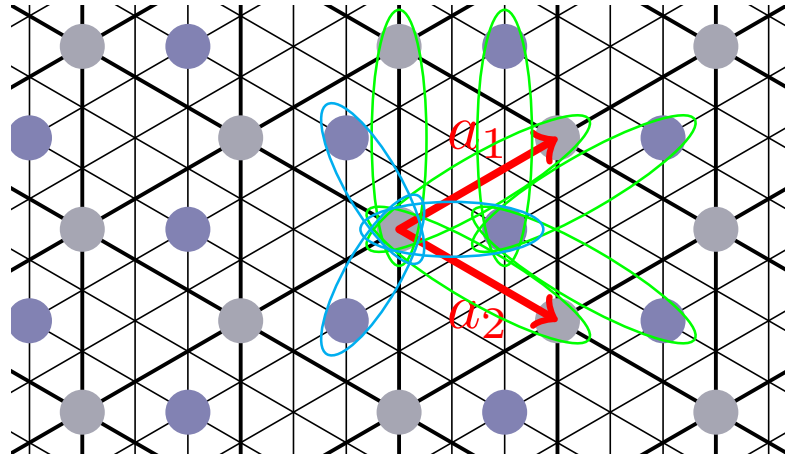


Figure 2: Dimers in graphene for nearest and next-nearest neighbor.

All translationally inequivalent pairs within nearest neighbor and next nearest neighbor have been illustrated; all others will be generated by the translation group. Each pair will have both a σ and a π type difference. Below I write out the potential:

$$\begin{aligned}
 V = & \frac{1}{2} \gamma_{\sigma} \sum_{\eta=0,1,2} \left(-\cos\left(\frac{2\pi}{3}\eta\right) u_{(i,j)}^{(A,x)} - \sin\left(\frac{2\pi}{3}\eta\right) u_{(i,j)}^{(A,y)} + \cos\left(\frac{2\pi}{3}\eta\right) u_{v_{\eta}^{i,j}}^{(B,x)} + \sin\left(\frac{2\pi}{3}\eta\right) u_{v_{\eta}^{i,j}}^{(B,y)} \right)^2 + \\
 & \frac{1}{2} \gamma_{\pi} \sum_{\eta=0,1,2} \left(\sin\left(\frac{2\pi}{3}\eta\right) u_{(i,j)}^{(A,x)} - \cos\left(\frac{2\pi}{3}\eta\right) u_{(i,j)}^{(A,y)} - \sin\left(\frac{2\pi}{3}\eta\right) u_{v_{\eta}^{i,j}}^{(B,x)} + \cos\left(\frac{2\pi}{3}\eta\right) u_{v_{\eta}^{i,j}}^{(B,y)} \right)^2 + \\
 & \frac{1}{2} \gamma'_{\sigma} \sum_{\alpha;\eta=0,1,2} \left(-\cos\left(\frac{\pi}{3}\eta + \frac{\pi}{6}\right) u_{(i,j)}^{(\alpha,x)} - \sin\left(\frac{\pi}{3}\eta + \frac{\pi}{6}\right) u_{(i,j)}^{(\alpha,y)} + \cos\left(\frac{\pi}{3}\eta + \frac{\pi}{6}\right) u_{w_{\eta}^{i,j}}^{(\alpha,x)} + \sin\left(\frac{\pi}{3}\eta + \frac{\pi}{6}\right) u_{w_{\eta}^{i,j}}^{(\alpha,y)} \right)^2 + \\
 & \frac{1}{2} \gamma'_{\pi} \sum_{\alpha;\eta=0,1,2} \left(\sin\left(\frac{\pi}{3}\eta + \frac{\pi}{6}\right) u_{(i,j)}^{(\alpha,x)} - \cos\left(\frac{\pi}{3}\eta + \frac{\pi}{6}\right) u_{(i,j)}^{(\alpha,y)} - \sin\left(\frac{\pi}{3}\eta + \frac{\pi}{6}\right) u_{w_{\eta}^{i,j}}^{(\alpha,x)} + \cos\left(\frac{\pi}{3}\eta + \frac{\pi}{6}\right) u_{w_{\eta}^{i,j}}^{(\alpha,y)} \right)^2
 \end{aligned}$$

η	0	1	2
$v_{\eta}^{i,j}$	(i, j)	$(i, j-1)$	$(i-1, j)$
$w_{\eta}^{i,j}$	$(i+1, j)$	$(i+1, j-1)$	$(i, j+1)$

The index α sums over both basis atoms within the primitive unit cell (ie. A and B). The tuples $v_{\eta}^{i,j}$ and $w_{\eta}^{i,j}$ take the following values:

η	0	1	2
$v_{\eta}^{i,j}$	(i, j)	$(i, j-1)$	$(i-1, j)$
$w_{\eta}^{i,j}$	$(i+1, j)$	$(i+1, j-1)$	$(i, j+1)$

You can find the coefficients in table II of Phys. Rev. B **67** 035401 (2003).

$$\gamma_{\sigma} = 27.3 \quad \gamma'_{\sigma} = 4.69 \quad \gamma_{\pi} = 10.205 \quad \gamma'_{\pi} = -2.615$$

These are in units of $\frac{eV}{A^2}$. In order to make the frequencies come out in cm^{-1} , then do the following:

```
[ np.sign(s)*np.sqrt(abs(s)/(12.011*1.66054*10**-27*6.2415*10**-2)))*6.582*10**-16*8065.6
  for s in np.linalg.eigh(temp)[0]]
```

(a) Compute the eigenvalues of the Hamiltonian along the path:

$$\Gamma \rightarrow M \rightarrow K \rightarrow \Gamma$$

In four different cases:

- Using only γ_{σ}
- Using γ_{σ} and γ_{π}
- Using γ_{σ} and γ_{π} and γ'_{σ}
- Using γ_{σ} and γ_{π} and γ'_{σ} and γ'_{π}

In each case, please do write out all non-zero force constant matrices. You should then be able to derive the analytic formula for the Dynamical matrix in each case. You can then use a computer to diagonalize the 4x4 matrix for any particular value of k that you like and plot the results. Here are the answers (top left, top right, bottom left, bottom right, respectively). Please see if you can obtain these curves:

